

3. Să se determine formula de interpolare de formă

$$\int_{-\infty}^{\infty} e^{-x^2} f(x) dx = A f(a) + R(f)$$

 care are gradul de exactitate maxim.
 Să se deducă formula restului și evoluția acestuia
 dacă $f \in C^2(\mathbb{R})$

$$R(e_0) = \int_{-\infty}^{\infty} e^{-x^2} dx - A = \sqrt{\pi} - A = 0$$

deci $A = \sqrt{\pi}$;

$$R(e_1) = \int_{-\infty}^{\infty} x e^{-x^2} dx - \sqrt{\pi} * a = 0$$

deci $a = 0$

$$R(e_2) = \int_{-\infty}^{\infty} x^2 e^{-x^2} dx = \frac{1}{2} \sqrt{\pi} \neq 0.$$

Gradul de exactitate e 1.

Formula restului:

$$R(f) = R(e_2) \frac{f''(c)}{2!}$$

4. Să se deducă o formulă de interpolare de
 grad de exactitate maxim de formă:

$$\int_a^b f(x) dx = A f(b) + B f(x_1) + R(f), \quad x_1 \in (a, b)$$

$$R(e_0) = \int_a^b dx - A - B = b - a - (A + B) = 0$$

$$R(e_1) = \int_a^b x dx - A * b - B * x_1 = 0 = -\frac{1}{2}a^2 + \frac{1}{2}b^2 - Ab - Bx_1$$

$$R(e_2) = \int_a^b x^2 dx - A * b^2 - B * x_1^2 = 0 = -\frac{1}{3}a^3 + \frac{1}{3}b^3 - Ab^2 - Bx_1^2$$

sistemul:

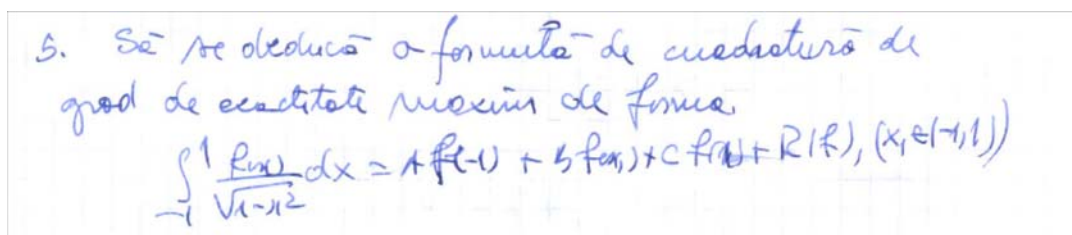
$$\begin{aligned} b - a - (A + B) &= 0 \\ -\frac{1}{2}a^2 + \frac{1}{2}b^2 - Ab - Bx_1 &= 0 \\ -\frac{1}{3}a^3 + \frac{1}{3}b^3 - Ab^2 - Bx_1^2 &= 0 \end{aligned}$$

, Solution is: $A = -\frac{1}{4}a + \frac{1}{4}b$, $B = -\frac{3}{4}a + \frac{3}{4}b$, $x_1 = \frac{2}{3}a + \frac{1}{3}b$
Formula:

$$\int_a^b f(x) dx = \frac{b-a}{4} \left(f(b) + 3f\left(\frac{2a+b}{3}\right) \right) + R(f).$$

$$R(e_3) = \int_a^b x^3 dx - \frac{b-a}{4} * b^3 - 3 * \frac{b-a}{4} \left(\frac{2a+b}{3} \right)^3 = -\frac{1}{36} (a-b)^4 \neq 0$$

:



$$\begin{aligned} R(e_0) &= \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} dx - A - B - C = 0 \\ \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} dx &\stackrel{x=\sin t}{=} \int_{-\pi/2}^{\pi/2} \frac{\cos t dt}{\sqrt{1-\sin^2 t}} = \pi \end{aligned}$$

rezultă:

$$A + B + C = \pi$$

$$R(e_1) = \int_{-1}^1 \frac{x}{\sqrt{1-x^2}} dx + A - Bx_1 - C = 0$$

rezultă:

$$A = C + Bx_1$$

$$R(e_2) = \int_{-1}^1 \frac{x^2}{\sqrt{1-x^2}} dx - A - Bx_1^2 - C = 0$$

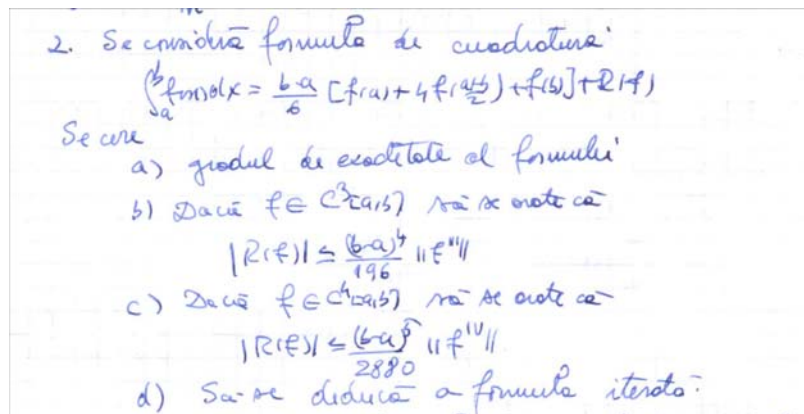


Figure 1:

$$-Bx_1^2 + \frac{1}{2}\pi - A - C = 0$$

$$R(e_3) = \int_{-1}^1 \frac{x^3}{\sqrt{1-x^2}} dx + A - Bx_1^3 - C = 0$$

Adică:

$$A - Bx_1^3 - C = 0$$

Sistemul:

$$\begin{aligned} A + B + C &= \pi \\ A &= C + Bx_1 \\ -Bx_1^2 + \frac{1}{2}\pi - A - C &= 0 \\ A - Bx_1^3 - C &= 0 \end{aligned}$$

, Solution is: $[A = \frac{1}{4}\pi, B = \frac{1}{2}\pi, C = \frac{1}{4}\pi, x_1 = 0]$

Gradul de exactitate este cel puțin 3. : Calculăm:

$$R(e_4) = \int_{-1}^1 \frac{x^4}{\sqrt{1-x^2}} dx - \frac{\pi}{4} \left((-1)^4 + 0 + 1 \right) = -\frac{1}{8}\pi \neq 0$$

gr. este 3. :

a) Calculez $R(e_k)$, $k=0,1,\dots$ până este nul.

$$\begin{aligned}
R(e_0) &= \int_a^b dx - \frac{b-a}{6} (1+4+1) = 0 \\
R(e_1) &= \int_a^b x dx - \frac{b-a}{6} \left(a + 4\frac{a+b}{2} + b \right) = 0 \\
R(e_2) &= \int_a^b x^2 dx - \frac{b-a}{6} \left(a^2 + 4\left(\frac{a+b}{2}\right)^2 + b^2 \right) = 0 \\
R(e_3) &= \int_a^b x^3 dx - \frac{b-a}{6} \left(a^3 + 4\left(\frac{a+b}{2}\right)^3 + b^3 \right) = 0 \\
R(e_4) &= \int_a^b x^4 dx - \frac{b-a}{6} \left(a^4 + 4\left(\frac{a+b}{2}\right)^4 + b^4 \right) = \frac{(a-b)^5}{120} \neq 0
\end{aligned}$$

deci gradul de exactitate este 3.

b. Formula, cf. T. este de tip interpolator:

$$\begin{aligned}
\int_a^b f(x) dx &= \int_a^b L(f; a, (a+b)/2, b)(x) dx + \int_a^b R_L(f)(x) dx = \\
&= \int_a^b \frac{(x - \frac{a+b}{2})(x-b)}{(a - \frac{a+b}{2})(a-b)} f(a) + \frac{(x-a)(x-b)}{(\frac{a+b}{2} - a)(\frac{a+b}{2} - b)} f\left(\frac{a+b}{2}\right) + \frac{(x-a)(x - \frac{a+b}{2})}{(b-a)(b - \frac{a+b}{2})} f(b) dx \\
\int_a^b \frac{(x - \frac{a+b}{2})(x-b)}{(a - \frac{a+b}{2})(a-b)} dx &= \frac{b-a}{6} = \int_a^b \frac{(x-a)(x - \frac{a+b}{2})}{(b-a)(b - \frac{a+b}{2})} dx \\
\int_a^b \frac{(x-a)(x-b)}{(\frac{a+b}{2} - a)(\frac{a+b}{2} - b)} dx &= 4\frac{b-a}{6}
\end{aligned}$$

$$\begin{aligned}
\left| \int_a^b R_L(f)(x) dx \right| &\leq \int_a^b |R_L(f)(x)| dx = \\
&= \int_a^b \left| (x-a) \left(x - \frac{a+b}{2} \right) (x-b) \right| \left| \frac{f'''(c_x)}{3!} \right| dx \leq \\
&\leq \frac{\|f'''\|}{3!} \int_a^b \left| (x-a) \left(x - \frac{a+b}{2} \right) (x-b) \right| dx = \\
&= \frac{\|f'''\|}{3!} \frac{(b-a)^4}{32} = \frac{(b-a)^4}{196} \|f'''\|.
\end{aligned}$$

c. Dacă formula e de tip interpolator și are gradul de exactitate m atunci:

$$R(f) = R(e_{m+1}) \frac{f^{(m+1)}(c)}{(m+1)!}$$

în cazul de mai sus $m = 3$ deci:

$$R(f) = R(e_4) \frac{f^{iv}(c)}{4!} = \frac{(a-b)^5}{120} \frac{f^{iv}(c)}{4!} = \frac{(a-b)^5}{2880} f^{iv}(c).$$

d. Formula iterată:

$$\int_a^b f(x) dx = \frac{b-a}{6n} \left(f(a) + f(b) + 2 \sum_{k=1}^{n-1} f\left(a + \frac{b-a}{n}k\right) + 4 \sum_{k=1}^{n-1} f\left(a + \frac{b-a}{2n}(2k-1)\right) \right) + R(f)$$

1. Fie $\Delta: 0 = x_0 < x_1 < \dots < x_n = 1$, $n \in \mathbb{N}$. Se cere să se arate că

$$\int_0^1 L_n^2(f; x_0, \dots, x_n)(x) dx \geq \sum_{k=0}^{n-1} \frac{(f(x_{k+1}) - f(x_k))^2}{x_{k+1} - x_k}$$

Ideea: se folosește teorema:

TEOREMĂ (Legea de minim pt $S_\Delta(f)$) Există o unică funcție $g^* \in \mathcal{F}$ astfel că

$$\inf_{g \in \mathcal{F}} \int_0^1 g'^2 dx = \int_0^1 g^{*2} dx$$

și $g^* = S_\Delta(f)$

Deoarece $L(f; x_0, \dots, x_n)(x_i) = f(x_i)$ avem g^* pt. L și f este aceeași. Atunci:

$$\begin{aligned} \int_0^1 L_n^2(f; x_0, \dots, x_n)(x) dx &\geq \int_0^1 g^{*2}(x) dx = \\ &= \sum_{k=0}^{n-1} \int_{x_k}^{x_{k+1}} g^{*2}(x) dx = \\ &= \sum_{k=0}^{n-1} \left(\frac{f(x_{k+1}) - f(x_k)}{x_{k+1} - x_k} \right)^2 \int_{x_k}^{x_{k+1}} dx \\ &= \sum_{k=0}^{n-1} \left(\frac{f(x_{k+1}) - f(x_k)}{x_{k+1} - x_k} \right)^2 (x_{k+1} - x_k). \end{aligned}$$

2. Fie $f \in C[0,1]$, $\Delta: 0 = x_0 < x_1 < \dots < x_n = 1$. Se cere să se arate că există $c \in [0,1]$ astfel că

$$\inf_{g \in \mathcal{F}} \int_0^1 g'^2 dx = f'(c)^2$$

Cf. teoremei amintite și problemei precedente:

$$\begin{aligned} \inf_{g \in \mathcal{F}} \int_0^1 g'^2(x) dx &= \sum_{k=0}^{n-1} \left(\frac{f(x_{k+1}) - f(x_k)}{x_{k+1} - x_k} \right)^2 (x_{k+1} - x_k) = \\ &= \sum_{k=0}^{n-1} \frac{(f'(c_k))^2 (x_{k+1} - x_k)}{\sum_{k=0}^{n-1} (x_{k+1} - x_k)} = (f'(c))^2. \end{aligned}$$

3. Să se arate că $\inf_{g \in \mathcal{F}_2} \int_0^1 g'^2(x) dx \geq \sum_{k=2}^n (x_{k+1} - x_k) [x_k, x_{k+1}]^2$

Ideea e aceeași ca la problema 1, folosind teorema corespunzătoare pt. splinuri cubice.

4. Să se calculeze.

$$\inf_{g \in \mathcal{F}_2} \int_0^1 g''^2(x) dx$$

Se aplică corolarul:

Avem:

$$\int_0^1 S''_{\Delta,3}(x) dx = \sum_{k=0}^{n-1} \int_{x_k}^{x_{k+1}} S''_{\Delta,3}(x) dx = ?$$

dar pe $[x_k, x_{k+1}]$ $S''_{\Delta,3}(x)$ este polinom de grad 1.

$$\begin{aligned} S'_{\Delta,3}(x_k) &= f'(x_k), S'_{\Delta,3}(x_{k+1}) = f'(x_{k+1}) \\ S_{\Delta,3}(x_k) &= f(x_k), S_{\Delta,3}(x_{k+1}) = f(x_{k+1}) \end{aligned}$$

COROLLARY

$$\inf_{g \in \mathcal{G}_2} \int_0^1 g''^2 dx = \int_0^1 S_{\Delta_3}''^2 dx.$$

Figure 2:

$$\begin{aligned} S''(x) &= A(x - x_k) + B(x - x_{k+1}) \\ S'(x) &= \frac{A}{2}(x - x_k)^2 + \frac{B}{2}(x - x_{k+1})^2 + C(x_{k+1} - x) + C(x - x_k) \\ \frac{A}{2}(x_{k+1} - x_k)^2 + C(x_{k+1} - x_k) &= f'(x_{k+1}) \\ \frac{B}{2}(x_k - x_{k+1})^2 + C(x_{k+1} - x_k) &= f'(x_k) \\ (A - B)(x_{k+1} - x_k)^2 &= f'(x_{k+1}) - f'(x_k) \end{aligned}$$